

THE MONOTONICITY THEOREM IN O-MINIMAL GEOMETRY

PÁL AMBRUS AND TEAM

Definition 0.1. A dense linear order without endpoints is an ordered pair $\mathcal{R} = (R, <)$, where R is a set and $<$ is a binary relation on R , satisfying the following axioms:

- (1) **Strict Linear Order:**
 - $\forall x \in L, \neg(x < x)$ (Irreflexivity)
 - $\forall x, y, z \in L, (x < y \wedge y < z) \rightarrow x < z$ (Transitivity)
 - $\forall x, y \in L$, exactly one of $x < y$, $y < x$, or $x = y$ holds (Trichotomy)
- (2) **Density:**
 - $\forall x, y \in L, (x < y) \rightarrow \exists z \in L, (x < z \wedge z < y)$
- (3) **No Endpoints:**
 - $\forall x \in L, \exists y \in L, (y < x)$ (No left endpoint)
 - $\forall x \in L, \exists y \in L, (x < y)$ (No right endpoint)

Definition 0.2. Let $\mathcal{R} = (R, <)$ be a dense linear order without endpoints. An **o-minimal structure** on R is a structure $\mathcal{S} = \{S_n\}_{n \in \mathbb{N}}$, where S_n is a collection of subsets of R^n satisfying the following axioms:

- (1) **Structure Axioms:**
 - (a) S_n is a Boolean subalgebra of $\mathcal{P}(R^n)$ (closed under union, intersection, complement, and contains the empty set).
 - (b) S_n contains all diagonal sets, i.e., sets of the form $\{(x_1, \dots, x_n) \in R^n : x_i = x_j\}$ for any $1 \leq i < j \leq n$.
 - (c) S_n is closed under Cartesian products ($A \in S_n, B \in S_m \implies A \times B \in S_{n+m}$).
 - (d) S_n is closed under coordinate projections ($A \in S_{n+1} \implies \pi(A) \in S_n$, where $\pi : R^{n+1} \rightarrow R^n$ is the projection on one of the coordinates).
 - (e) The relation $<$ is in S_2 (i.e., the set $\{(x, y) \in R^2 : x < y\}$ is in S_2).
- (2) **O-minimality Axiom:** The set S_1 consists of exactly the finite unions of points and open intervals $\{x \in R : a < x < b\}$, where $a, b \in R \cup \{-\infty, +\infty\}$.

We say that a subset of R^n is definable (with respect to $\mathcal{S}$) if it is an element of S_n .

Definition 0.3. Let $\mathcal{M} = (R, <, \{S_n\}_{n \in \mathbb{N}})$ be an o-minimal structure. Let $A \subseteq R^m$ and $B \subseteq R^n$ be definable. A function $f : A \rightarrow B$ is said to be a **definable function** if its graph,

$$\Gamma(f) = \{(\bar{x}, f(\bar{x})) \in R^m \times R^n : \bar{x} \in A\},$$

is a definable subset of R^{m+n} .

Theorem 0.4 (Monotonicity Theorem). *Let $f : (a, b) \rightarrow R$ be a definable function on the interval (a, b) . Then there are points*

$$a = a_0 < a_1 < \dots < a_k = b$$

such that on each subinterval (a_i, a_{i+1}) , the function is either constant, or strictly monotone and continuous.

Proof. We derive this from the three lemmas below. In these lemmas we consider a definable function $f : I \rightarrow R$ on an interval I .

Lemma 0.5. *There is a subinterval of I on which f is constant or injective.*

Lemma 0.6. *If f is injective, then f is strictly monotone on a subinterval of I .*

Lemma 0.7. *If f is strictly monotone, then f is continuous on a subinterval of I .*

These lemmas imply the monotonicity theorem as follows: let

$$X := \{x \in (a, b) \mid \text{on some subinterval of } (a, b) \text{ containing } x \text{ the function } f \\ \text{is either constant, or strictly monotone and continuous}\}.$$

Now $(a, b) - X$ must be finite, since otherwise it would contain an interval I ; applying successively Lemmas 1, 2, and 3 we can make I so small that f is either constant, or strictly monotone and continuous, on I . But then $I \subseteq X$, a contradiction. Since $(a, b) - X$ is finite, we can reduce the proof of the theorem to the case that $(a, b) = X$, by replacing (a, b) by each of the finitely many intervals of which the open set X consists. In particular, we may assume that f is continuous. By splitting up (a, b) further we can reduce to one of the following three cases.

Case 1. For all $x \in (a, b)$, f is constant on some neighborhood of x .

Case 2. For all $x \in (a, b)$, f is strictly increasing on some neighborhood of x .

Case 3. For all $x \in (a, b)$, f is strictly decreasing on some neighborhood of x .

Case 1. Take $x_0 \in (a, b)$ and put

$$s := \sup\{x : x_0 < x < b, f \text{ is constant on } [x_0, x)\}.$$

Then $s = b$, since $s < b$ implies that f is constant on some neighborhood of s , contradiction. From $s = b$ it follows that f is constant on $[x_0, b)$. Similarly, we prove that f is constant on $(a, x_0]$. Therefore f is constant on (a, b) .

Case 2. Take $x_0 \in (a, b)$ and put

$$s := \sup\{x : x_0 < x < b, f \text{ is strictly increasing on } [x_0, x)\}.$$

Then $s = b$, since $s < b$ leads to a contradiction as in case 1. Therefore f is strictly increasing on $[x_0, b)$. Similarly, f is strictly increasing on $(a, x_0]$. Hence f is strictly increasing on (a, b) .

Case 3. This is handled in the same way as case 2.

We now prove the lemmas.

Proof of Lemma 1. If some $y \in R$ had infinite preimage $f^{-1}(y)$, then this preimage would contain a subinterval of I and f would take the constant value y on that subinterval. So we may assume that each $y \in R$ has finite preimage. Then $f(I)$ is infinite, and so contains an interval J . Define an “inverse” $g : J \rightarrow I$ by

$$g(y) := \min\{x \in I : f(x) = y\}.$$

Since g is injective by definition, $g(J)$ is infinite, and hence $g(J)$ contains a subinterval of I , and f is necessarily injective on this subinterval. $\square$

Proof of Lemma 2. Let us write $I = (a, b)$. We assume here that f is injective and have to show that f is strictly monotone on some subinterval of I . For each $x \in I$ the interval (a, x) is a disjoint union of two subsets,

$$(a, x) = \{y \in (a, x) : f(y) < f(x)\} \cup \{y \in (a, x) : f(y) > f(x)\},$$

so one of the parts contains an interval (c, x) , $a < c < x$. The interval (x, b) breaks up similarly. This shows that each $x \in I$ satisfies exactly one of the following four formulas:

$$\Phi_{++}(x) := \exists c_1, c_2 \in I [c_1 < x < c_2 \& \forall y \in (c_1, x) : f(y) > f(x) \& \forall y \in (x, c_2) : f(y) > f(x)],$$

$$\Phi_{+-}(x) := \exists c_1, c_2 \in I [c_1 < x < c_2 \& \forall y \in (c_1, x) : f(y) > f(x) \& \forall y \in (x, c_2) : f(y) < f(x)],$$

and $\Phi_{-+}(x)$ and $\Phi_{--}(x)$, which are defined similarly.

So I contains a subinterval all of whose points satisfy the same formula. Replacing I by this subinterval, and f by its restriction to that subinterval, we may assume that all points satisfy the same formula. This leads to four cases.

EASY CASE. $\Phi_{-+}(x)$ for all x in I .

For each x in I define $s(x) := \sup\{s \in (x, b) : f > f(x) \text{ on } (x, s)\}$. Then clearly $s(x) = b$, since $s(x) < b$ contradicts $\Phi_{-+}(s(x))$. Therefore f is strictly increasing on I .

The case that $\Phi_{+-}(x)$ for all x in I leads similarly to the conclusion that f is strictly decreasing on I .

DIFFICULT CASE. $\Phi_{++}(x)$ for all x in I .

Let $B := \{x \in I : \forall y \in I (y > x \text{ implies } f(y) > f(x))\}$. If B is infinite then B contains an interval, and on this interval f is strictly increasing, and we are done. So let us assume that B is finite. Passing to a subinterval to the right of all points of B we may assume

$$(*) \quad \forall x \in I \exists y \in I (y > x \& f(y) < f(x)).$$

Let $c \in I$. We claim that for all large enough y in I we have $f(y) < f(c)$. Otherwise, we would have $f(y) > f(c)$ for all large enough $y \in (c, b)$. Take then $d \in [c, b)$ minimal such that $\forall y (d < y < b \text{ implies } f(y) > f(c))$.

If $f(d) > f(c)$ then d would not be minimal since $\Phi_{++}(d)$. So $f(d) < f(c)$. But by $(*)$, there is e with $d < e < b$ and $f(e) < f(d)$, so $f(e) < f(c)$, contradiction. This proves the claim that $f(y) < f(c)$ for all large enough $y \in I$.

Define $y(c)$ as the least element of $[c, b)$ for which $f(y) < f(c)$ if $y(c) < y < b$. Note that $\Phi_{++}(c)$ gives $c < y(c)$ and $f(y(c)) < f(c)$. Minimality of $y(c)$ clearly implies that $y(c)$ satisfies the following formula $\Psi_{+-}(v)$:

$$\Psi_{+-}(v) := \exists v_1, v_2 \in I [v_1 < v < v_2 \& \forall z_1, z_2 (v_1 < z_1 < v < z_2 < v_2 \rightarrow f(z_1) > f(z_2))].$$

Since c was arbitrary we have shown $\forall c \in I \exists v \in I (v > c \& \Psi_{+-}(v))$.

Therefore $\Psi_{+-}(v)$ holds for all v in an interval of the form (d, b) , $d \in I$. Replacing I by this subinterval we may as well assume that $\Psi_{+-}(v)$ holds on all of I .

A completely similar argument shows that we can pass to a still smaller subinterval on which Ψ_{-+} (defined in the obvious way) holds. But this is a contradiction since we cannot simultaneously have $\Psi_{+-}(v)$ and $\Psi_{-+}(v)$. The case that $\Phi_{--}(x)$ holds for all x in I is completely similar to the case we just handled. This finishes the proof of lemma 2. $\square$

Proof of Lemma 3. Let us assume that the strictly monotone function f is strictly increasing. (The case that f is strictly decreasing goes the same way.) Since $f(I)$ is infinite there is an interval $J \subseteq f(I)$. Take two points $r, s \in J$, $r < s$, and let c, d be their preimages: $f(c) = r$, $f(d) = s$, $c < d$. Clearly f defines an order preserving bijection of (c, d) onto (r, s) . But the topology is defined in terms of the order, hence f is continuous on (c, d) . $\square$

This finishes the proof of the lemmas and hence the proof of the monotonicity theorem is complete. $\square$

GEOMETRIA TANSZÉK, EÖLTVÖS LORÁND TUDOMÁNYEGYETEM, 1117 BUDAPEST, PÁZMÁNY PÉTER SÉTÁNY 1/C

E-mail address: ambrus.pal@elte.ttk.hu